

MINGLANG YIN, Ph.D.

myin16@jhu.edu | Website: minglangyin.github.io | GitHub: github.com/MinglangYin | [Google Scholar](#)
AI/ML · Modeling · Statistics · Representation Learning · High-Performance Computing

SUMMARY

- ML/AI researcher with 10+ years modeling sparse, noisy, high-dimensional real-world data.
- Depth in predictive/data-driven modeling, representation learning, foundation models, survival analysis, and computational science.
- Hands-on high-performance computing and distributed GPU training.
- 20+ peer-reviewed AI/ML publications in *Nature-family*, *AAAI*, and *leading applied mathematics venues*.
- Motivated to apply ML/AI research to quantitative finance.

EDUCATION

Brown University Providence, RI
Ph.D., Engineering (Biomedical) and **M.Sc.**, Engineering (Fluid and Thermal Sciences) 2016 – 2022
Advisor: Prof. George Em Karniadakis (Division of Applied Mathematics)

Northwestern Polytechnical University Xi'an, China
B.S., Engineering (Aeronautical) 2012 – 2016

TECHNICAL SKILLS

Programming: Python, PyTorch, C++.

Statistics and Modeling: Bayesian inference, survival analysis, Markov Chain Monte Carlo (MCMC); gradient boosting (XGBoost/LightGBM), transformer, time series forecasting, numerical PDEs, numerical methods.

High-Performance Computing: distributed multi-GPU training, OpenMPI.

RESEARCH EXPERIENCE

Johns Hopkins University Baltimore, MD
Senior Postdoctoral Researcher & AI Group Lead Aug 2022 – Present
Advisors: Profs. Natalia Trayanova (Biomedical), Mauro Maggioni (Applied Mathematics & Statistics)

Published in Nature-Family journals and top CS, medical venues: *Nature Computational Science*, *Nature Cardiovascular Research*, *AAAI*, *Journal of the American Heart Association*, *JACC: Clinical Electrophysiology*, and *Computers in Biology and Medicine*; presented in *Heart Rhythm Scientific Sessions* and *Society for Industrial and Applied Mathematics (SIAM)*. Two manuscripts under review.

Time-Series Foundation Model & Predictive Modeling

- Proposed a novel AI framework for predicting spatiotemporal system response on geometries and achieved 10,000× speedup over numerical solvers; accomplished 98% accuracy on terabytes of real-world data.
- Developed a Transformer-based Time-Series Foundation Model; validated on >350 TB of realistic data with accuracy >95%.
- Expedited training using distributed multi-GPU training, Flash Attention, and Group-query attention; enhanced model generalizability on out-of-distribution (OOD) data with modern LLM/NLP techniques: RoPE, Mixture of Experts.

Multimodal AI for Forecasting & Survival Analysis

- Developed a multimodal AI framework fusing images with tabular covariates to predict time-to-event outcomes and forecast risk for multiple scenarios, outperforming the established benchmarks by 15–35%; validated on held-out external datasets to quantify robustness under distribution shift.
- Led the end-to-end study design, data processing, and ideation; built the full statistical stack — hypothesis testing, feature selection, survival analysis, and model evaluation and calibration — and deployed the model externally at a partner institution.

Feature Extraction, Representation Learning on High-Dimensional Data

- Developed a dimensionality reduction method for high-dimensional geometric data, improved forecasting baselines with the low-dimensional representations by 16% and identified drivers of outcome.
- Secured competitive external grant funding (2-year \$100K), led end-to-end study design, data curation and preprocessing, statistical analysis, large-scale computation, and visualization.

Brown University

Ph.D. Researcher, Division of Applied Mathematics

Advisor: Profs. George Karniadakis (Applied Mathematics & Engineering)

Providence, RI

2018 – 2022

Published in top applied mathematics, CS conference, and computational venues: *Computer Methods in Applied Mechanics and Engineering*, *Journal of the Royal Society Interface*, *AAAI*, *Journal of the Mechanics and Physics of Solids*, *Computational Mechanics*, and *Acta Mechanica Sinica*; presented at *American Physical Society* and *SIAM* conferences.

Bayesian Inference & Uncertainty Quantification

- Developed a generative ML model coupled with Markov Chain Monte Carlo for parameter estimation and uncertainty quantification on sparse, experimental data; outperformed Gaussian process regression and nonlinear least-squares baselines with a 50% reduction in relative error.
- Built physics-informed ML models to infer latent parameters from sparse, noisy observations for ill-posed problems.

High-Performance Computing

- Implemented a parallel computing solver module in a production-level C++ codebase (LAMMPS).
- Designed an ML–numerical solver coupling scheme achieving $>14,000\times$ speedup over direct simulation.
- Implemented, and validated a new solver module for a 3D parallel PDE solver (Nektar).
- Co-designed a hybrid GPU–CPU pipeline for multiscale simulation, demonstrating linear scaling.

Surrogate Modeling for Nonlinear Systems

- Built reduced-order models to approximate expensive high-fidelity fluid and solid simulations at a fraction of the computational cost.
- Developed data-driven surrogates for forecasting nonlinear dynamical systems from simulations and observational data.

SELECTED PUBLICATIONS

Survival Analysis, Statistics, Multimodal AI

- **M. Yin** et al. Predicting the *When*: Multimodal AI for Time-to-Recurrence Analysis After Atrial Fibrillation Ablation, *Journal of the American Heart Association*, 2026.
- C. Lai, **M. Yin** et al. Multimodal AI to forecast arrhythmic death in hypertrophic cardiomyopathy. *Nature Cardiovascular Research*, 2025. [[JHU Hub](#), [Johns Hopkins Medicine](#), [New York Magazine](#)]

Predictive Modeling, Surrogate Modeling

- **M. Yin** et al. A scalable framework for learning geometry-dependent solution operators of PDEs. *Nature Computational Science*, 2024. [[99th percentile](#); [Communication of the ACM](#), [MIT Tech Review](#), [NSF News](#), [JHU Hub](#)]
- E. Zhang, **M. Yin**, et al. Physics-informed neural networks for nonhomogeneous material identification in elasticity imaging, *AAAI Conference*, 2020

Markov Chain Monte Carlo, Generative, Uncertainty Quantification

- **M. Yin**, et al. A generative modeling framework for inferring families of biomechanical constitutive laws in data-sparse regimes. *Journal of the Mechanics and Physics of Solids*, 2023.

SELECTED GRANTS & AWARDS

- | | |
|--|------|
| ○ AHA Rapid Impact Research Award | 2026 |
| ○ Best PhD Thesis Award (Int. J. Numer. Methods Biomed. Eng.) | 2024 |
| ○ Kenneth M. Rosen Fellowship | 2023 |
| ○ Chinese Government Award for Outstanding PhD Students Abroad (Top 1/600 globally) | 2022 |